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Palaeomagnetism and Magnetic-Polarity Zonation in some Oligocene Volcanic Rocks of the San Juan Mountains, South-western Colorado

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Summary

Palaeomagnetic results have been obtained from thirty sites in intrusive and extrusive rocks of Oligocene age from the San Juan Mountains, south-western Colorado. All specimens from each site were subjected to af demagnetization, and the reliability of each site determined. Twenty-three sites gave reliable results. Because five sites from the thick intracaldera part of the La Jara Canyon Member of the Treasure Mountain Tuff appear to have become magnetized during the same small interval of geological time, their results were combined and their mean pole and direction used in the final calculations. The eighteen remaining reliable sites yielded an average Oligocene palaeomagnetic pole at 85·6° N and 298·0° E ($\delta p = 8°$, $\delta m = 11°$, $k = 24 \cdot 1$). A stratigraphic sequence is given for the major San Juan ash-flow sheets and selected additional units with their corresponding magnetic polarities and mean K–Ar ages.

Introduction

There have been few palaeomagnetic investigations of mid-Tertiary North American rocks. The Oligocene volcanics of the San Juan Mountains in south-western Colorado are very suitable for palaeomagnetic work but the only previous palaeomagnetic study of these rocks has been field magnetic-polarity determinations by P. W. Lipman on the major ash-flow sheets (Lipman & Steven 1970; Popenoe & Steven 1969). The work presented here is the result of a palaeomagnetic reconnaissance of some of the major volcanic and hypabyssal units of the San Juan Mountains with emphasis on the Platoro caldera complex (Lipman & Steven 1970; Lipman 1974, in press) in an attempt to obtain a palaeomagnetic pole for the Oligocene and thus define a point on the North American polar wandering curve for the mid-Tertiary.

The volcanic rocks of the San Juan Mountains belong to the San Juan volcanic field, which covers about 25 000 km^2 in Colorado and adjacent parts of New Mexico and constitutes the largest erosional remanent of a once nearly continuous volcanic field that covered much of the southern Rocky Mountains and adjacent areas during Oligocene and later time (Steven & Epis 1968). Recent studies by Lipman, Steven & Menhert (1970) using potassium–argon dating have greatly modified the volcanic

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history for the region as originally worked out by Larson & Cross (1956). This new data shows that initial volcanism in the San Juan field consisted of alkali andesite, rhyodacite, and mafic quartz latite lavas and breccias from local volcanic centres. Isotopic dates indicate that most of these intermediate rocks formed during the interval from 35 to 30 My ago. Approximately 30 My ago, eruption of intermediate lavas gave way to voluminous ash-flow eruptions of quartz latite and low-silica rhyolite, in a phase that ended about 26 My ago. These ash-flow sheets are genetically related to large caldera complexes in the central and western San Juan Mountains. Interlayered with and locally overlying these ash-flow sheets are later lavas and breccias of intermediate composition. During this period a few epizonal porphyritic stocks, dikes, and laccoliths intrude the volcanic pile. These hyabyssal intrusives are mainly monzonites and granodiorites and yield radiometric ages ranging from 26 to 30 My. The Alamosa River stock, the largest of these exposed intrusive bodies, has been dated at $29{\cdot}1 \pm 1{\cdot}2$ My (Lipman *et al.* 1970). During the early Miocene, volcanism changed from the predominantly intermediate lavas and silicic differentiates to a bimodal suite of basalt and rhyolite. Basalt and small quantities of rhyolite were erupted intermittently throughout the Miocene and Pliocene, with some flows as young as Pleistocene. These lavas once formed a thin cover over much of the older volcanic field, but now are highly eroded.

Sampling and measurements

Fig. 1 shows the distribution of sampling sites. The bulk of the sampling sites are located within units related to the composite Platoro caldera complex. This

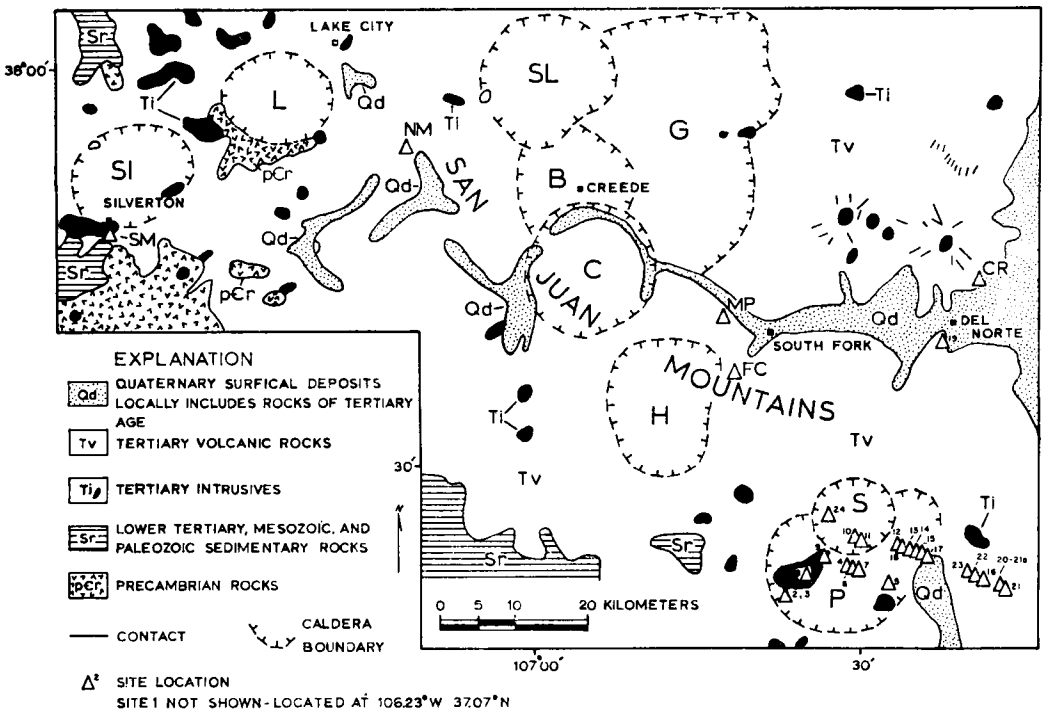


FIG. 1. Geologic map of the San Juan Mountains area (generalized from Larson & Cross 1956; Lipman & Steven 1970) showing site locations and calderas (P, Platoro caldera; S, Summitville caldera; C, Creede caldera; B, Bachelor caldera; G, La Garita caldera; SL, San Luis caldera; L, Lake City caldera; SI, Silverton caldera).

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complex includes the Platoro and Summitville calderas, features that subsidided recurrently during eruption of ash-flows of the upper Oligocene Treasure Mountain Tuff (Lipman 1974, in press). In addition, one site each was located in younger ash-flow sheets related to the Mt Hope (Masonic Park Tuff), La Garita (Fish Canyon Tuff), Bachelor (Carpenter Ridge Tuff), and Cochetopa (Nelson Mountain Tuff) calderas (Lipman *et al.* 1970). One other site was located in the Sultan Mountain stock near Silverton. Due to considerations of accessibility and weathering, most sampling was limited to artificial exposures in roadcuts. From 4 to 10 samples were drilled and oriented '*in situ*', using a gasoline powered drill similar to that described by Doell & Cox (1967) and a magnetic compass. Back azimuths were taken to insure against orientation error caused by local magnetic anomalies. Samples collected at each site were spread over distances ranging from 25 to 100 m. At four sites (20, 21a, 23, 24) only one or two samples were collected due to sampling problems. These sites were used for polarity determination only and have not been included in the directional calculations reported below.

Remanent magnetism was measured with a Schonstedt SM-1A spinner magnetometer. All samples were magnetically cleaned by alternating field demagnetization, using an apparatus fitted with a four-axis tumbler. The two specimens, from each site showing the greatest difference in NRM direction were used as pilot samples for demagnetization. These pilot samples were demagnetized in increments of either 50 or 100 Oe starting at either 50 or 100 Oe until the field that produced the smallest angular difference was obtained. This field was then used to demagnetize all other specimens from the site. Optimum demagnetization fields ranged from 100 to 1200 Oe. Specimens that appeared to diverge from a sampling group at the optimum demagnetization level were demagnetized at higher fields until they reached a stable end point. Most specimens responded well to magnetic cleaning, but the directions of a few specimens changed erratically upon repeated demagnetization and measurement, indicating that they probably contained little or no stable remanence. In a few other specimens the presumed NRM direction could not be recovered, probably due to its having been replaced by a strong secondary component of magnetization acquired during recent times. This condition was indicated by directions that were approximately aligned with the Earth's present field direction before and after demagnetization, or that strongly diverged from the remainder of their group. Both varieties of presumably unreliable specimens were eliminated from further considerations. Remanent directions of these discarded specimens (15 in all) were at least 50 degrees from the remainder, with the discrepancy usually being much greater.

Rarely in palaeomagnetic research do all sites yield results of sufficient accuracy to constitute reliable estimates of the ancient geomagnetic field direction. Irving (1964, p. 102) suggests as minimum standards for a reliable site at least five specimens cut from individually oriented cores yielding a mean direction that has a circle of 95 per cent confidence of less than 25° radius. Beck (1972) uses four independent specimens, with a circle of confidence of less than 15°; Creer (1962) states that only four or five specimens will give a statistically valid direction, and Wilson (1970) uses circles of 95 per cent confidence of less than 12°. All of these criteria are, of course, completely arbitrary. For the present study, in order to minimize effects of dispersion related to instability or other extraneous conditions, the following criteria were established to determine the reliability of a site:

(1) An acceptable site must have at least four specimens cut from individually oriented samples.

(2) The circle of 95 per cent confidence for the site must be less than 20°.

Four sites (20, 21a, 23, 24) were eliminated due to criterion 1, and three sites (11, 13, 17) were eliminated due to criterion 2. Twenty-three sites survived these rejection

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Table 1

Summary of palaeomagnetic data for sites meeting reliability criteria, San Juan Mountains, Colorado

Unit	S	N	D	I	k	α_{95}	Demagnetization level (Oe)	Pole location N° lat.	Pole location E° long.
La Jara Canyon Member of the Treasure Mtn Tuff	1	7	204·2	−43·6	133·3	5·2	100	66·5	6·5
	16	8	200·9	−42·2	99·1	5·6	200	68·0	13·5
	LJC*	2	202·5	−42·9	683·1	6·1		67·0	10·0
	2	8	178·5	−37·7	75·4	6·4	100	74·0	78·5
	3	7	193·7	−37·6	210·4	4·2	100	70·0	33·5
	4	7	187·3	−43·7	148·0	5·0	600	76·5	43·5
	6	4	195·4	−32·3	27·3	17·9	1200	66·0	35·0
	7	5	180·7	−40·0	137·3	6·6	1000	75·5	71·0
	LJC**	5	187·3	−38·5	124·0	6·9		73·0	49·5
	12	7	137·0	−63·4	209·7	4·2	200	57·0	190·5
	18	7	178·8	−63·6	203·0	4·2		82·0	247·5
Alamosa River Stock	8	8	39·2	66·3	191·9	4·0	250	59·5	309·0
	9	7	235·5	−72·4	174·4	4·6	150	49·0	295·5
Summitville And. Upper Member	10	4	169·2	−48·1	37·4	15·2	300	78·0	124·0
Summitville And. Lower Member	5	6	182·0	−78·9	36·8	11·2	300	58·5	255·0
Summitville And. Lower Member	14	6	180·5	−47·7	18·2	16·2	1000	81·5	71·0
	15	5	216·3	−53·0	28·8	14·5	400	60·5	340·0
Conejos Fm	19	4	198·9	−52·0	132·4	8·0	400	74·0	356·0
	22	5	173·8	−53·2	122·2	7·0	600	84·0	130·0
Ojito Creek Member of the Treasure Mtn Tuff	21	8	332·8	48·3	269·2	3·4	200	66·0	152·0
Sultan Mtn Stock	SM	10	357·7	63·4	260·2	3·0	100	82·5	239·5
Fish Canyon Tuff	FC	10	9·2	45·8	258·2	3·0	100	77·0	34·0
Nelson Mtn Tuff	NM	7	355·8	67·5	319·6	3·4	100	69·0	206·5
Masonic Pk Tuff	MP	10	206·8	−70·4	230·8	3·2	400	65·0	292·0
Carpenter Ridge Tuff	CR	9	165·4	−63·1	390·6	2·6	200	77·0	200·0

Explanation: *S*, site designation; *N*, number of samples; *D* and *I*, mean declination and inclination of site; *k* and α_{95} (Fisher 1953) statistical parameters; LJC* average for sites 1 and 16 and LJC** average for sites 2, 3, 4, 6, and 7, see discussion.

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criteria and are listed in Table 1, corrected for geological dip. The largest dip correction was 15°; most were in the range of 5°.

Angular dispersion and secular variation

Angular dispersion of site-mean directions from all stable sites is about 16°. This dispersion could be the result of experimental error associated with field and laboratory procedures, or from inaccurate dip corrections. However, we feel these factors have been minimized and contribute only one or two degrees to the total angular dispersion of the site mean directions. If this dispersion is then attributable to geomagnetic secular variation, it is in good agreement with studies by Creer & Sanver (1969), which show that geomagnetic secular variation during the Late Tertiary and Quaternary was slightly greater on the average than predicted by the Creer–Cox model (Creer 1962; Cox 1962). Secular variation during the sampled portion of the Oligocene for the San Juan region, therefore, probably was similar to that during the Late Tertiary and Quaternary.

Nine out of the 30 sites drilled were located within the La Jara Canyon Member of the Treasure Mountain Tuff; two in the widespread outflow part of the ash-flow sheet and seven in the thick intracaldera part. Mean directions of the two sites located in the relatively thin outflow sheet are nearly identical, probably because they acquired their magnetization at nearly the same time. For this reason, these two sites were averaged together and given unit weight in final directional calculations. However, the thick intracaldera tuff probably cooled more slowly, thereby acquiring its magnetization at a different time from the quicker cooling ash-flow sheet. Fig. 2 is a plot of mean directions from sites located within the thick intracaldera tuff. Mean directions from sites 12 and 18 obviously diverge significantly from the other five sites, which are well grouped. Explanations based on undetected post-magnetization tilting appear unlikely; more probably this divergence reflects a different magnetization time. Since different parts of the formation (which has a thickness of roughly 100 m outside the caldera and 800 m inside) probably became magnetized at different times, secular variation of the ancient geomagnetic field must have produced a certain amount of scatter in directions of magnetization. The La Jara Canyon Member is multiple flow cooling unit, and sites 12 and 18 come from the

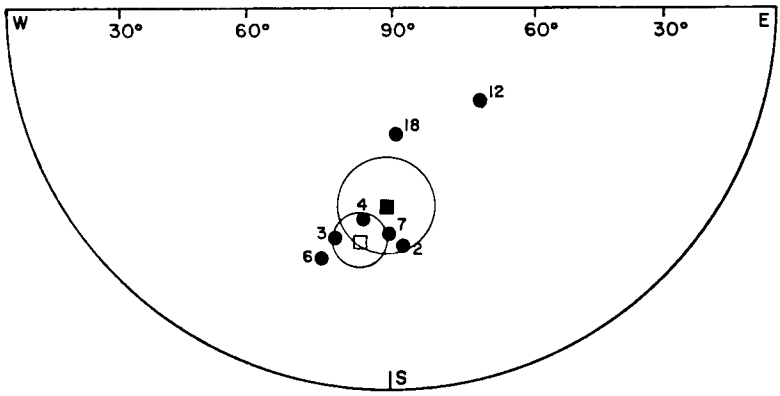


FIG. 2. Equal-area plot of the site-mean directions for the thick intracaldera part of the La Jara Canyon Member of the Treasure Mountain Tuff. All points are in the upper hemisphere. Solid box is the mean direction of remanent magnetization for all intracaldera La Jara Canyon sites. Open box is the mean direction of remanent magnetization from the La Jara Canyon Member sites 2, 3, 4, 6, and 7. Circles represent 95 per cent confidence limits.

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uppermost part of the thick intracaldera tuff. The other five sites are stratigraphically lower. Therefore, it is possible that sites 12 and 18 simply acquired TRM at a time that was significantly different than the magnetization time of the other five sites. Alternatively, as the upper part of the intracaldera tuff is not densely welded, it may have been relatively more permeable and therefore subject to chemical modification during subsequent intracaldera volcanic activity. Thus, the divergence of sites 12 and 18 could also be the result of remagnetization by CRM, probably during a somewhat later stage of Platoro volcanic activity. In either case, sites 12 and 18 were probably magnetized, wholly or partly, at a time different from the other five sites.

Since the divergence of sites 12 and 18 of the intracaldera tuff may be the result of the time differences in magnetization and since the other five sites show a very tight grouping, it is probable that the latter sites acquired their magnetization at much the same point in geological time. To eliminate any bias in calculating the mean pole for the San Juan volcanics that might be introduced by giving unit weight to each of the five closely grouped intracaldera site-mean directions, a mean direction and pole have been calculated for the five and given unit weight, along with sites 12 and 18 (which probably represent different points in time), in the pole calculation.

Polarity and palaeomagnetic pole position

Radiometric dates indicate that the volcanic and hyabyssal units sampled cover nearly the entire range of Oligocene time. It seems reasonable therefore to assume

Table 2

Mean palaeomagnetic pole

	N	*R*	*k*	δ	δ*p*	δ*m*	N° lat.	E° long.
All sites	18	17·2938	24·07	16·1	8·2°	10·8°	85·6	298·0

Explanation: *N*, number of sites; *R*, *k*, (Fisher 1953) statistical parameters; δ, angular dispersion (Wilson 1959); δ*p*, δ*m* are semi-axes of the oval of 95 per cent confidence about the palaeomagnetic pole (Irving 1964, p. 69).

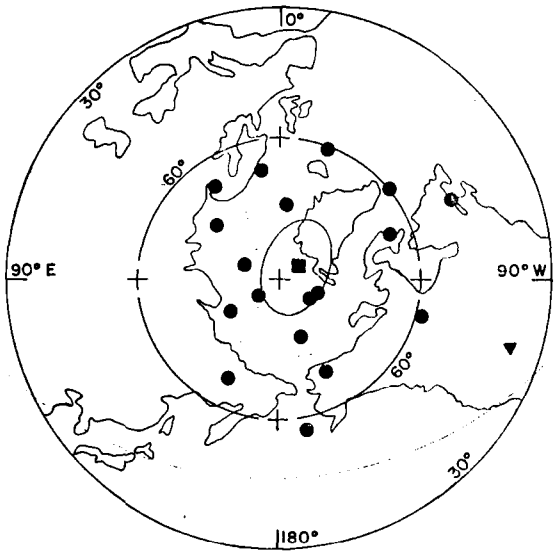


FIG. 3. Polar equal-area projection showing site-mean poles (after grouping of La Jara Canyon Member sites) and the mean palaeomagnetic pole (square) calculated for the San Juan region with its oval of 95 per cent confidence. Δ is sampling location.

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that the calculated mean pole position averages over enough time to eliminate secular variation effects and give a reliable estimate of the palaeomagnetic pole position for the Oligocene. The palaeomagnetic pole position presented here has been calculated from the mean of the site-mean directions for all sites meeting the reliability criteria discussed earlier. The pole and associated dispersion statistics are listed in Table 2. As seen in Fig. 3, the mean pole position of all sites is not statistically different from the present geographic pole at the 95 per cent level. If Wilson's (1970; 1971) dipole-offset hypothesis is correct, the true position of the palaeogeographic pole for the period studied is located even farther south (nearer Colorado) than calculated; the San Juan pole then might be statistically distinct from the present geographic pole. The data presented here cannot be used to test Wilson's hypothesis, except in conjunction with data from other continents. Even so, the effects of post-Oligocene continental drift would have to be eliminated before a rigorous test could be made. Such an attempt is not made in this study.

Comparison of the pole position reported above with other mid-Tertiary pole positions from North America is made in Table 3 and in Fig. 4. Overlapping circles of 95 per cent confidence indicate that the San Juan pole and the pole from the Spanish Peaks dike swarm (Larson & Strangway 1969) of late Oligocene–early Miocene age (Stormer 1962; Smith 1973) are not significantly different and probably represent much the same position of the mid-Tertiary palaeomagnetic pole. How-

Table 3

Middle-Tertiary pole positions, North America

	Lat °N	Long °E	δM	δP	α_{95}
Spanish Peaks, Colorado (Larson & Strangway 1969)	81·0	211·0	20	16	—
Marys Peak sill, Oregon (Clark 1969)	63·0	196·0	10	6	—
Volcanics, Western U.S. (Grommé & McKee 1971)	78·0	146·0	—	—	5·8
San Juan Mtns, Colorado (this study)	85·6	298·0	11	8	—

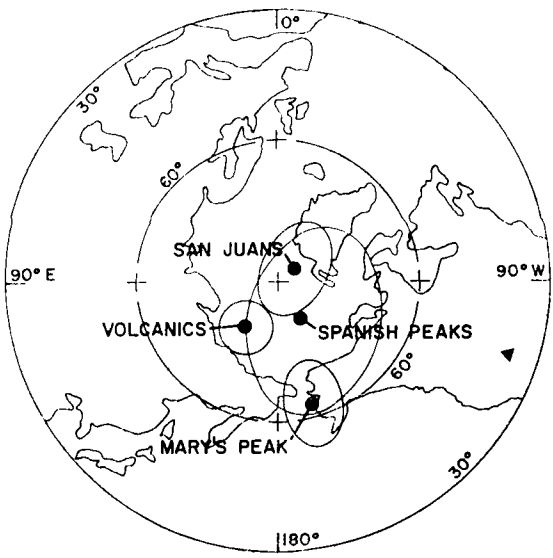


FIG. 4. Polar equal-area projection comparing the Oligiocene San Juan pole with other mid-Tertiary poles from North America. Δ is sampling loaction.

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Table 4

Stratigraphic sequence, polarity, and mean K–Ar ages of the major San Juan ash-flow sheets and selected additional units

Unit	Caldera	Mean age (My)	Polarity
Sultan Mountain Stock (SM)	Silverton	25·5	*N*
Fisher Quartz Latite	Creede	26·4	*N*
Snowshoe Mountain Tuff	Creede	26·4	*N*
Nelson Mountain Tuff (NM)	Cochetopa		*N*
Rat Creek Tuff	San Luis		*N*
Wason Park Tuff	(near Creede)		*R*
Mammouth Mountain Tuff	(near Creede)	26·7	*R*
Carpenter Ridge Tuff (CR)	Bachelor		*R*
Fish Canyon Tuff (FC)	La Garita	27·8	*N*
Masonic Park Tuff (MP)	Mt. Hope	28·2	*R*
Alamosa River Stock (8, 9)		29·1	*N*-core *R*-margin
Upper Member, Summitville Andesite (10, 11, 24)	Summitville		*R*
Ra Jadero Member, Treasure Mtn Tuff (20)	Summitville		*R*
Ojito Creek Member, Treasure Mtn Tuff (21, 21a)	Summitville		*N*
Lower Member, Summitville Andesite (5, 13, 14, 15, 17)	Summitville		*R*
La Jara Canyon Member, Treasure Mtn Tuff (1, 2, 3, 4, 6, 7, 12, 16, 18)	Platoro	29·8	*R*
Conejos Formation		30–35	*N* & *R*

Explanation: *N*(*R*) is normal (reversed) polarity; ( ) sampling sites that are located within the named unit. K–Ar ages from Lipman and others (1970).

ever, circles (ovals) of 95 per cent confidence from Grommé & McKee's (1971) pole for volcanic rocks (22–37 My from Utah, Nevada, Oregon, and California and the San Juan pole just fail to overlap. The cause of this divergence is unknown, but could be the result of time differences in the intervals represented by the two poles. In contrast, the pole from the Marys Peak sill ($29{\cdot}7 \pm 1{\cdot}2$ My) in Oregon (Clark 1969) is far removed from the San Juan pole. Clark estimates that the sill took between 2000 and 6000 years to cool. Palaeomagnetic data from the Late Tertiary and Quaternary indicate that the effects of secular variation will average out over a period of 10^4 years (Creer 1962). A pole calculated from a rock unit representing less than 10^4 years may not have averaged out the effects of secular variation and therefore may not accurately represent the true ancient position of the geographic pole. Thus, it would seem reasonable to assume that the Marys Peak pole does not represent a true mid-Tertiary palaeogeographic pole.

The laboratory-determined magnetic polarities reported here (Table 4) confirm in all cases preliminary magnetic polarities of the major ash-flow sheets of the San Juan field (Lipman & Steven 1970; Popenoe & Steven 1969) that were previously determined in the field by P. W. Lipman using a portable flux-gate magnetometer (Doell & Cox 1962). These field determinations on numerous samples from a single outcrop and from widely separated outcrops of the same unit were consistent for the major ash-flow sheets, except where obviously disrupted by lightning strikes. In contrast, samples of intermediate-composition lavas and intrusives commonly yielded inconsistent field measurements, probably as a result of secondary magnetization. Accordingly, in Table 4 we present a stratigraphic sequence of the major San Juan ash-flow sheets and selected additional units, showing their mean K–Ar ages and preliminary magnetic polarities based both on field and laboratory determinations. Data from Table 4 in combination with data from Table 1 for the two sites located within the Alamosa River stock (anti-parallelism of directions, opposite polarities, and site location within stock) suggest that the stock has cooled through and recorded a reversal of the Earth's magnetic field.

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Conclusions

Eighteen reliable sites from the volcanic and hypabyssal units in the San Juan Mountains have yielded an average Oligocene pole at 85·6° N and 298·0° E. The angular dispersion of the site-mean directions from these rocks fits well with observational studies of secular variation for the Late Tertiary and Quaternary. The stratigraphic sequence given for the major San Juan units with their magnetic polarities and mean K–Ar ages will perhaps be an aid for future mapping in this region.

The work presented here is only a reconnaissance of some of the major volcanic and hyabyssal units in the San Juan Mountains. The above results indicate that these rocks are very suitable for palaeomagnetic analysis. Additional palaeomagnetic data from all units within the San Juan region could lead to a precisely defined and well dated point on the curve of apparent polar wandering for North America for use in tectonic analysis and continental reconstructions.

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